

assignment 1

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1 a lidl text

here we note our findings while playing with tsetlin automata. they are pretty neat. i used google searches and the built in LLM and this [1] to figure out the code

2 Changing the number of states

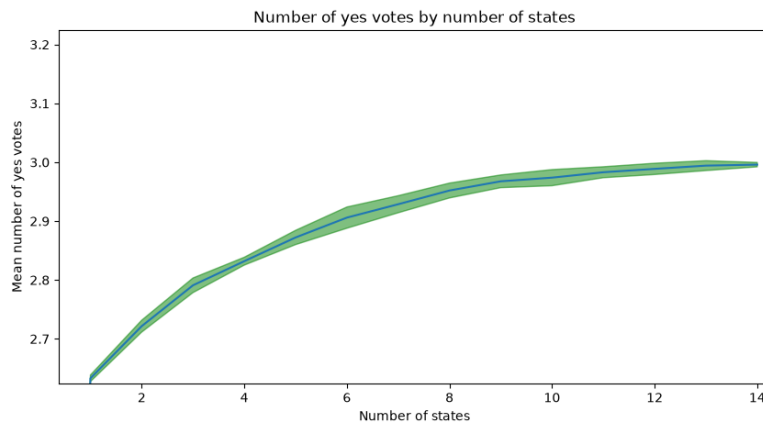


Figure 1:

Even at only 2 states the automata are nearly able to average 3 votes over 10000 iterations. by adding more depth/levels to the automata system performance improves quickly. we ran this 10 times and note bigger uncertainty(shaded area) at even numbers. this is because you're not supposed to use even numbers since it can cause weird behaviour depending on implementation, but it was minor as seen here so we didn't bother too much.

3 changing the target

we changed the target amount of votes and correspondingly change the code for calculating the rewards c_1 and c_2 as shown in code snippet in figure 2. It works well for a target of 4 as seen in fig. 3, but for target 2 as shown in fig.4

```
def reward(self, m, correct_m=4): 1 usage (1 dynamic)
# 0 or 1 or 2 or 3
if m <= correct_m:
    reward = 1 if random.random() < m * 0.2 else -1
    #p=0 for m=0, p=0.2 for m=1, p=0.4 for m=2, p=0.6 for m=3
# 4 and 5
else :
    reward = 1 if random.random() < ((correct_m*0.2)-((m-correct_m)*0.2)) else -1
    #p = 0.4 for m=4, p=0.2 for m=5
self.update(reward)
#print(f'a{self.name} reward is {reward} for m = {m}')
return reward
```

Figure 2: works for 4, but not for 2

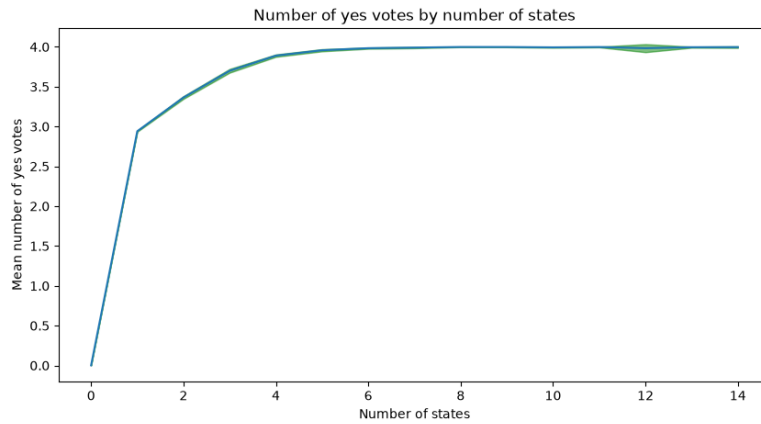


Figure 3: it works!

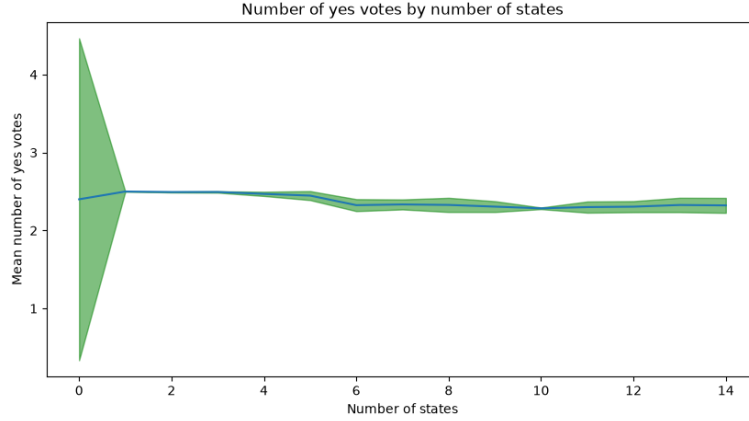


Figure 4: it fails for 2

4 more automata

we try increasing the number of automata. here all automata only have a depth of 3 and system performance is quickly negatively affected. the average number of yes votes increases for higher numbers of players first slowly then suddenly.

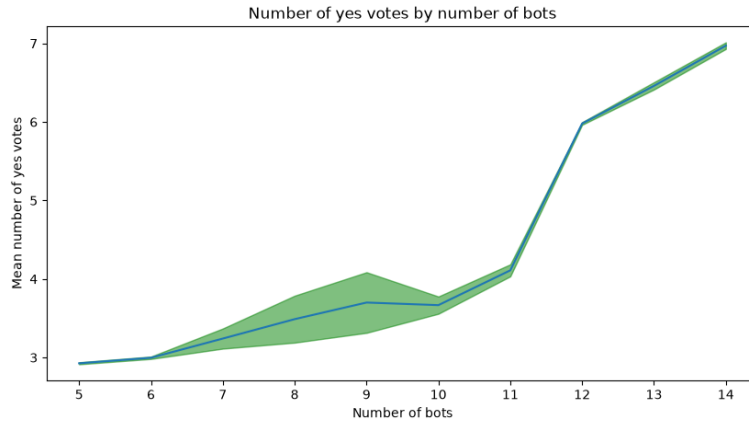


Figure 5: just increasing the number of automata hurts performance

If we increase the number of players they only need slightly more depth to average 3 yes votes per round over 10000 iterations. From 2 we found that performance stopped improving noticeably after giving automata seven levels of deepness. We found that increasing the number of players works fine as long as you increase the number of levels. we also tried putting in much bigger numbers

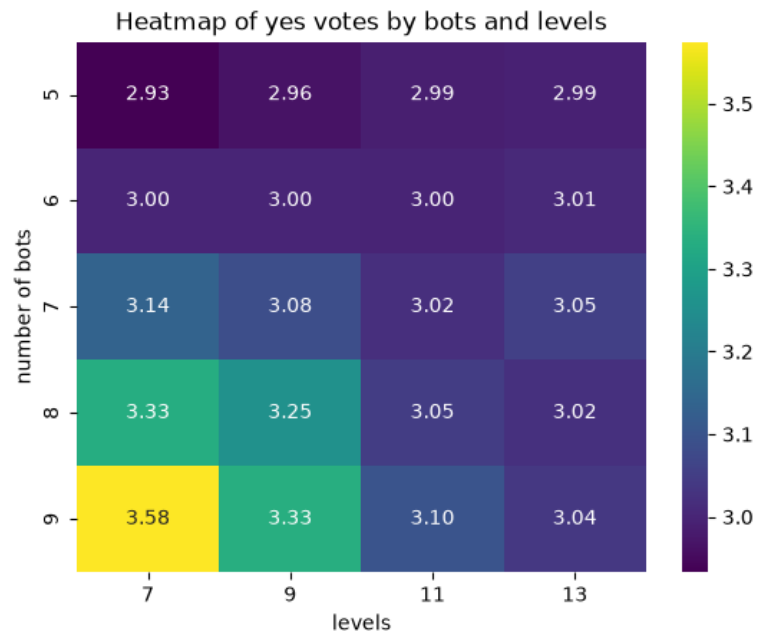


Figure 6: increasing the number of bots works fine as long as you increase the levels

and found that making the target to hit 4 made a

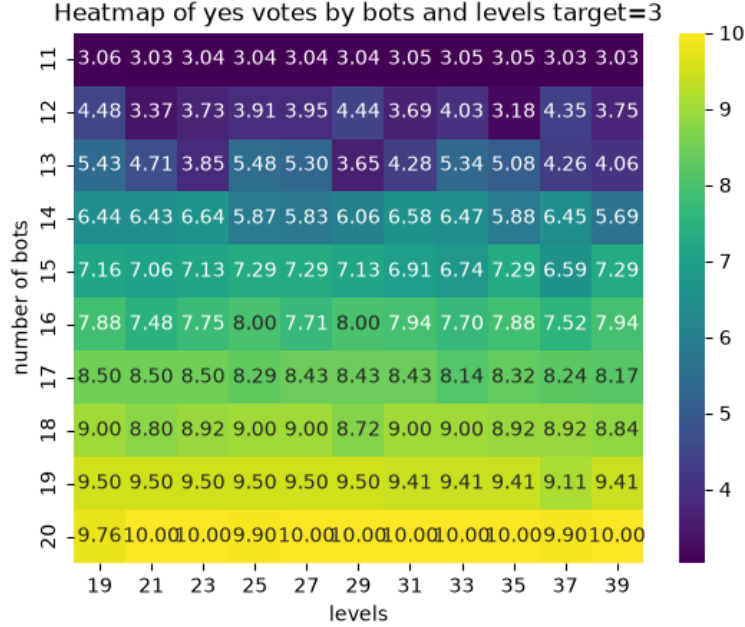


Figure 7: Too many players!

5 notes at the end

we find that making the problem harder by making there be more players competing and reducing their odds of beating the game by random selection can be solved by making there be more levels of depth. when we increase the target to 4 we can increase the number of players a lot and still have good performance.

6 code

code is at <https://github.com/haakbl99/TsetlinStuff>

References

- [1] Siddhartha kar. “Tsetlin Machine”. In: *Medium blog* (2024). URL: <https://medium.com/@siddharthakar24/tsetlin-machine-231c31f1f3a7>.

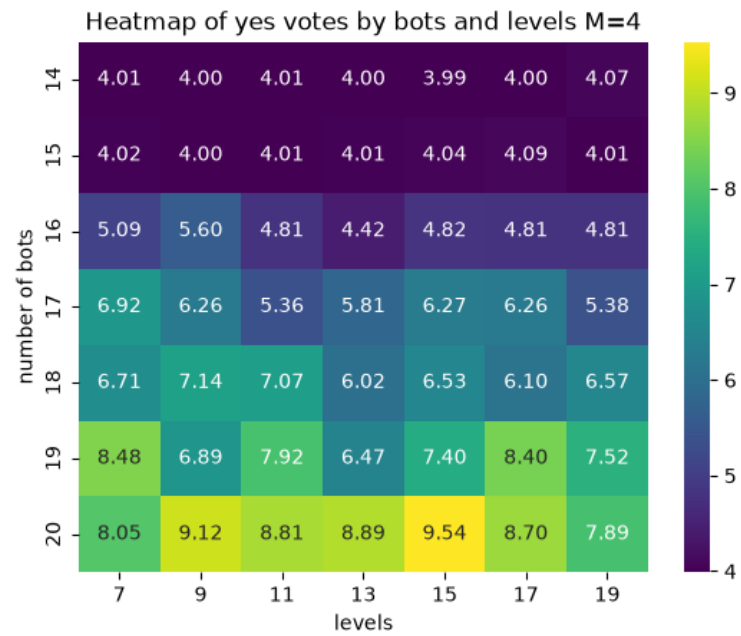


Figure 8: it's easier when more people can vote yes